Physics 201: Classical Dynamics Homework 4

Problem Statement

PenPress is an autonomous system built to print academic assignments, notes, and technical reports in natural handwriting using an Ender 3 Pro pen plotter. The software translates digital text or equations into continuous vector strokes with single line fonts. Our compiler schedules rapid travel moves with positive vertical clearance, eliminates all hotend and heatbed commands, and outputs standard dialect G-code for Marlin firmware. Before any physical motor moves, the virtual simulator executes the trajectory step by step, tracking machine coordinates, enforcing bed boundaries, and calculating trapezoidal velocity profiles. Finally, an optical recognition model reads the simulated toolpath render to verify that every single sentence is crisp, legible, and authentic. This closed loop pipeline guarantees reliable output on paper without human intervention.

Energy Conservation Equation:

Equation: `335995 E = rac{1}{2} m v^2 + V(x) = ext{const} 335995`